A PROJECT TITLE ON

Crop Recommendation Using Multinomial Logistic Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Name of Title: Crop Recommendation Using Multinomial Logistic Regression

1. Abstract:

Agriculture plays a vital role in the economy and food security of a country, and selecting a suitable crop for cultivation is one of the most important decisions made by farmers. Traditional crop selection often depends on experience, local practices, and weather conditions, which may not always provide accurate recommendations. This project, "Crop Recommendation Using Multinomial Logistic Regression," proposes a machine learning-based system that recommends the most suitable crop based on important agricultural and environmental parameters.

The proposed system uses Multinomial Logistic Regression, a supervised machine learning classification algorithm capable of predicting one class among multiple possible categories. The model takes input parameters such as nitrogen, phosphorus, potassium, temperature, humidity, pH value, and rainfall. These parameters are processed and provided to the trained machine learning model, which predicts the crop that is most suitable for the given conditions.

The system aims to assist farmers and agricultural users in making data-driven crop selection decisions. By analyzing the relationship between soil properties and environmental conditions, the model can provide a quick and systematic crop recommendation. The project also demonstrates the practical application of machine learning in agriculture and how classification techniques can be used to solve real-world problems.

2. INTRODUCTION:

Crop selection is an important part of successful agricultural production. Different crops require different soil nutrients, temperature, humidity, rainfall, and pH conditions for healthy growth. Choosing an unsuitable crop can result in reduced productivity and financial losses. Therefore, a system that can analyze agricultural conditions and recommend an appropriate crop can be useful for improving decision-making.

Machine learning provides techniques for analyzing large amounts of agricultural data and identifying patterns that may not be easily recognized through traditional methods. In this project, Multinomial Logistic Regression is used as a classification algorithm to predict the most suitable crop from multiple crop categories.

The system accepts agricultural parameters such as nitrogen, phosphorus, potassium, temperature, humidity, pH, and rainfall as input. The model is trained using a dataset containing different agricultural conditions and their corresponding suitable crops.The main purpose of this project is to demonstrate how machine learning can be applied to agriculture for making crop selection more systematic, efficient, and data-driven. It can serve as a basic decision-support system for farmers, students, researchers, and other agricultural users.

3. SOFTWARE REQUIREMENTS:

Programming Language
- Java (DSA Implementation)
- Python (Elastic Net Regression Model)

IDE
• IntelliJ IDEA / Eclipse (Java)
• Visual Studio Code / Jupyter Notebook (Python)
Libraries
Java
• File Handling
• Swing/JavaFX (Optional GUI)
Python
• NumPy
• Pandas
• Scikit-learn
• Matplotlib • Seaborn
Database
• CSV Dataset / MySQL (Optional)
Operating System
• Windows 10/11
• Linux
• macOS
Dataset
• TMDB Movie Dataset
• IMDb Movie Dataset
• Kaggle Movie Revenue Dataset
Hardware Requirements
• Intel Core i5 or above
• 8 GB RAM (minimum)
• 10 GB Free Storage

Signature of the Guide **Course Instructor**